10 Two market researchers, Riley and Ryan, are commissioned by a supermarket to investigate the total amount spent by shoppers in a single transaction.

Riley is instructed to collect data from 25 shoppers aged 18–39 and 25 shoppers aged 40–65.

Ryan is instructed to collect data from 30 shoppers aged 40–65 and 20 shoppers aged over 65.

(a) State the sampling method Riley and Ryan are using. **[1]**

(b) Explain whether this sampling method is random. **[1]**

The mean amount spent for each group is shown in the table.

	Riley's data		Ryan's data	
Age	18–39	40–65	40–65	over 65
Sample Size	25	25	30	20
Mean Amount Spent	£93.21	£38.22	£87.34	£65.22

(c) Find the mean of all the 100 transactions. **[2]**

A member of the supermarket management team states that one of the mean amounts for the 40–65 age group must be wrong.

(d) Explain why there is insufficient evidence to support this statement. **[1]**

10(a)

10(b)

10(c)

10(d)